

Solubility Prediction Step 4

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Importing libraries:

```
library(tidyverse)
library(kableExtra)
library("ggpubr")
library(plotly)
```

Reading the data file;

```
T1 <- read.csv("../data/results_df_2.csv", sep = ",", header = TRUE)
```

Seeing the structure of the data file:

```
str(T1, vec.len=2)

## 'data.frame': 7000 obs. of 10 variables:
## $ Model : chr "lr" "lr" ...
## $ Run : int 1 2 3 4 5 ...
## $ Train.MSE : num 1.28 1.28 ...
## $ Train.R2 : num 0.662 0.662 ...
## $ Train.RMSE: num 1.13 1.13 ...
## $ Train.MAPE: num 0.98 0.98 ...
## $ Test.MSE : num 1.05 1.05 ...
## $ Test.R2 : num 0.765 0.765 ...
## $ Test.RMSE : num 1.03 1.03 ...
## $ Test.MAPE : num 0.766 0.766 ...
```

Subsetting the ET model:

```
T1_et <- subset(T1, Model == "et")
```

Seeing the structure of the ET model results:

```
str(T1_et, vec.len=2)

## 'data.frame': 1000 obs. of 10 variables:
## $ Model : chr "et" "et" ...
## $ Run : int 1 2 3 4 5 ...
## $ Train.MSE : num 0.000289 0.000289 ...
## $ Train.R2 : num 1 1 ...
## $ Train.RMSE: num 0.017 0.017 ...
## $ Train.MAPE: num 0.000422 0.000422 ...
## $ Test.MSE : num 0.693 0.694 ...
## $ Test.R2 : num 0.845 0.845 ...
## $ Test.RMSE : num 0.832 0.833 ...
## $ Test.MAPE : num 0.579 0.565 ...
```

Visualizing the results for ET:

```
fig1 <- T1_et %>%
  plot_ly(
    y = ~Test.MSE,
    type = 'violin',
    box = list(
      visible = T
    ),
    meanline = list(
      visible = T
    ),
    color = I("#241468"),
    x0 = 'Extra Trees Regreesion'
  )
```

```
fig1 <- fig1 %>%
  layout(
    yaxis = list(
      title = "Test MSE",
      zeroline = F
    )
  )
```

```
fig2 <- T1_et %>%
  plot_ly(
    y = ~Test.R2,
    type = 'violin',
    box = list(
      visible = T
    ),
    meanline = list(
      visible = T
    ),
    color = I("#9F0D7F"),
    x0 = 'Extra Trees Regreesion'
  )
```

```
fig2 <- fig2 %>%
  layout(
    yaxis = list(
      title = "Test R2",
      zeroline = F
    )
  )
```

```
fig3 <- T1_et %>%
  plot_ly(
    y = ~Test.RMSE,
    type = 'violin',
    box = list(
      visible = T
    ),
    meanline = list(
      visible = T
    ),
    color = I("#EA1179"),
```

```

    x0 = 'Extra Trees Regreesion'
  )
fig3 <- fig3 %>%
  layout(
    yaxis = list(
      title = "Test RMSE",
      zeroline = F
    )
  )
fig4 <- T1_et %>%
  plot_ly(
    y = ~Test.MAPE,
    type = 'violin',
    box = list(
      visible = T
    ),
    meanline = list(
      visible = T
    ),
    color = I("#F79BD3"),
    x0 = 'Extra Trees Regreesion'
  )
fig4 <- fig4 %>%
  layout(
    yaxis = list(
      title = "Test MAPE",
      zeroline = F
    )
  )
#creating subplot
fig <- subplot(fig1, fig2, fig3, fig4, nrows = 2, titleY = TRUE, titleX = TRUE, margin =
0.1 )
fig <- fig %>%layout(title = 'Extra Trees Regreesion 1000 runs')

# Update title
annotations = list(
  list(
    x = 0.2,
    y = 0.98,
    text = "MSE",
    xref = "paper",
    yref = "paper",
    xanchor = "center",
    yanchor = "bottom",
    showarrow = FALSE
  ),
  list(
    x = 0.8,
    y = 0.98,
    text = "R2",
    xref = "paper",

```

```

yref = "paper",
xanchor = "center",
yanchor = "bottom",
showarrow = FALSE
),
list(
  x = 0.2,
  y = 0.4,
  text = "RMSE",
  xref = "paper",
  yref = "paper",
  xanchor = "center",
  yanchor = "bottom",
  showarrow = FALSE
),
list(
  x = 0.8,
  y = 0.4,
  text = "MAPE",
  xref = "paper",
  yref = "paper",
  xanchor = "center",
  yanchor = "bottom",
  showarrow = FALSE
))

```

```
fig <- fig %>%layout(annotations = annotations)
```

```
fig
```

